Projective Entropy and the Emergence of Newtonian Gravity

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Abstract

We investigate a minimal operator-theoretic mechanism by which a Newtonian interaction may emerge from the variation of a projective entropy functional. For a positive elliptic operator A in three spatial dimensions, we define $S_\Pi[A] = \frac{1}{2}\log\det\{\}'A$ and show that, in the weak-perturbation regime, its first variation reduces to a resolvent trace, $\delta S_\Pi = \frac{1}{2}\mathrm{Tr}(A^{-1}\delta A)$. The spatial structure of the entropy response is therefore governed by the Green kernel of A.

Since the Green function of a Laplace-type operator in three dimensions decays as $1/r$, the entropy variation inherits a Newtonian long-range behavior. We demonstrate this mechanism numerically using a discrete three-dimensional Laplacian with localized symmetric perturbations. The simulations confirm three key features: (i) the validity of the first-order resolvent expansion, (ii) the emergence of a $1/r$ Green profile, and (iii) the existence of a finite spatial support threshold for generating a macroscopic amplitude shift. Compact perturbations modify the spectrum locally without producing a measurable long-range effect, whereas sufficiently extended perturbations induce a coherent Newtonian response.

The results provide operator-theoretic evidence that a Newtonian interaction can arise as a resolvent-mediated consequence of projective entropy variation, without assuming a fundamental force law. The analysis is restricted to the weak-field regime and does not address relativistic dynamics.

Keywords: Newtonian Gravity, Projective Entropy, Emergence

Contents

1 Introduction

Gravity is traditionally described as a geometric phenomenon. In general relativity, the gravitational interaction is encoded in the curvature of spacetime, and the Newtonian potential emerges as a weak-field limit of the Einstein equations. This geometric formulation has been extraordinarily successful, yet it leaves open the conceptual question of whether gravitational interaction must be fundamental, or whether it could arise as an effective response of a more primitive structure.

In parallel, operator-theoretic and spectral viewpoints have increasingly clarified the role of elliptic operators in encoding geometric and physical information. In particular, functional determinants and resolvents of Laplace-type operators play a central role in quantum field theory, statistical mechanics, and spectral geometry. Variations of log-determinants are known to generate non-local responses through the inverse operator, suggesting a possible bridge between entropy-like functionals and long-range interactions.

The purpose of this work is to explore a minimal mechanism by which a Newtonian interaction may emerge from the variation of a projective entropy functional. We consider a positive elliptic operator A acting on a three-dimensional domain and define an entropy functional proportional to the logarithm of its (regularized) determinant. We show that, in the weak-perturbation regime, the variation of this functional reduces to a resolvent trace,

$$\delta S_\Pi \sim \tfrac{1}{2}\mathrm{Tr}(A^{-1}\delta A), \tag{1}$$

so that the Green kernel of A governs the spatial structure of the response.

Since the Green function of a Laplace-type operator in three dimensions decays as $1/r$, the resulting entropy variation inherits a Newtonian kernel. This mechanism does not assume a fundamental force law. Instead, long-range interaction arises as a non-local consequence of the resolvent structure of the operator.

We further demonstrate numerically that the emergence of a macroscopic $1/r$ amplitude shift depends on the spatial support of the perturbation. Compact perturbations confined to a sufficiently small region modify the entropy locally without generating a measurable long-range effect. By contrast, perturbations extending over a finite spatial region induce a coherent modification of the Green amplitude consistent with a Newtonian response.

The results presented here provide operator-theoretic evidence that a gravitational interaction can arise as a resolvent-mediated effect of projective entropy variation. Functional determinants and spectral invariants play a central role in quantum geometry and effective field theory [1, 2]. The analysis is intentionally minimal and does not rely on additional geometric assumptions. Its scope is restricted to the Newtonian regime, leaving relativistic extensions for future investigation.

2 Projective Entropy Functional

2.1 Elliptic Operator and Regularized Determinant

Let A be a positive, self-adjoint elliptic operator of Laplace type acting on functions over a bounded three-dimensional domain Ω, with appropriate boundary conditions ensuring a discrete spectrum. We denote its eigenvalues by $\{\lambda_k\}_{k\geq 0}$, ordered non-decreasingly.

If A admits a zero mode, we define the pseudo-determinant

$$\det\{\}'A = \prod_{\lambda_k>0} \lambda_k, \tag{2}$$

excluding the vanishing eigenvalue. More generally, the determinant is understood in the regularized sense, for instance via zeta regularization,

$$\log\det\{\}'A = -\left.\frac{d}{ds}\right|_{s=0} \sum_{\lambda_k>0} \lambda_k^{-s}. \tag{3}$$

Regularized determinants of elliptic operators are well established in spectral geometry and quantum field theory [1, 3, 4].

We define the projective entropy functional by

$$S_\Pi[A] = \frac{1}{2}\log\det\{\}'A. \tag{4}$$

The normalization factor $\frac{1}{2}$ is conventional and will simplify subsequent expressions.

2.2 First Variation and Resolvent Trace

Consider a perturbation of the operator,

$$A \longrightarrow A + \delta A, \tag{5}$$

with δA symmetric and sufficiently small in operator norm.

Using the identity

$$\delta \log \det A = \mathrm{Tr}(A^{-1}\delta A), \tag{6}$$

valid for invertible operators, and extending it to the pseudo-determinant by restriction to the positive spectral subspace, the first variation of S_Π reads

$$\delta S_\Pi = \frac{1}{2}\mathrm{Tr}(A^{-1}\delta A). \tag{7}$$

This identity is standard in operator theory and spectral analysis [3, 4].

Thus, in the weak-perturbation regime, the entropy variation is governed by the resolvent of the operator. Denoting by $G(x, y)$ the Green kernel of A,

$$A^{-1}(x, y) = G(x, y), \tag{8}$$

the variation can be expressed formally as

$$\delta S_\Pi = \frac{1}{2}\int_\Omega \int_\Omega G(x, y)\, \delta A(y, x)\, dx\, dy. \tag{9}$$

The relation between resolvents, Green kernels, and spectral invariants is well established in the mathematical physics literature [4].

This expression makes explicit that the response induced by a local perturbation δA is mediated non-locally by the Green kernel. The spatial decay properties of $G(x, y)$ therefore determine the long-range structure of the entropy variation.

3 Green Kernel in Three Dimensions

3.1 Laplace-Type Operators and Asymptotic Behavior

We consider an elliptic operator of Laplace type in three spatial dimensions,

$$A = -\nabla^2 + V(x), \tag{10}$$

acting on a bounded domain $\Omega \subset \mathbb{R}^3$ with boundary conditions ensuring self-adjointness and positivity.

In the weak-field regime and away from boundaries, the leading-order behavior of the Green kernel is governed by the principal symbol of the operator. Neglecting lower-order contributions, the Green function satisfies

$$-\nabla^2 G(x, y) = \delta(x - y). \tag{11}$$

In three dimensions, the fundamental solution of the Laplacian is well known:

$$G(x,y) = \frac{1}{4\pi|x-y|}. \tag{12}$$

The relation between resolvents, heat kernels, and spectral invariants has been extensively studied in the mathematical physics literature [3, 4].

More generally, for operators of the form $A = -\nabla^2 + V(x)$ with sufficiently regular and bounded V, the Green kernel admits the asymptotic expansion

$$G(x,y) \sim \frac{1}{4\pi|x-y|} \quad \text{as } |x-y| \to \infty, \tag{13}$$

up to exponentially suppressed or subleading corrections, provided no mass gap is introduced.

3.2 Entropy Variation and Newtonian Kernel

Substituting this asymptotic form into the entropy variation formula,

$$\delta S_\Pi = \frac{1}{2} \int_\Omega \int_\Omega G(x,y)\, \delta A(y,x)\, dx\, dy, \tag{14}$$

we obtain, in the long-distance regime,

$$\delta S_\Pi \sim \frac{1}{8\pi} \int_\Omega \int_\Omega \frac{\delta A(y,x)}{|x-y|}\, dx\, dy. \tag{15}$$

If the perturbation δA is spatially localized, this expression reduces asymptotically to a convolution with the Newtonian kernel $1/|x-y|$. The entropy response induced by a compact perturbation therefore decays as $1/r$ at large distances.

The appearance of the Newtonian kernel is thus not imposed as a fundamental interaction law. It follows directly from the resolvent structure of a three-dimensional Laplace-type operator. Long-range behavior emerges from the spectral properties of the operator governing the entropy functional.

4 Numerical Evidence for Resolvent-Mediated Newtonian Response

4.1 Discrete Model

We approximate a three-dimensional Laplace-type operator by the graph Laplacian L on a cubic lattice with Dirichlet boundary conditions. The operator used in the simulations is

$$A = L + \varepsilon I, \tag{16}$$

with $\varepsilon > 0$ small enough to remove the zero mode while preserving the long-distance behavior of the Green kernel.

A localized perturbation is introduced by symmetrically modifying edge weights inside a ball of lattice radius R. The perturbed operator reads

$$A \longrightarrow A + \delta A, \tag{17}$$

with δA compactly supported.

The Green column associated with a point source is computed by solving

$$A\phi = \delta_{\text{source}}, \tag{18}$$

and the radial profile of ϕ is fitted at large distance by

$$\phi(r) \sim \frac{a}{r} + b. \tag{19}$$

The coefficient a characterizes the effective Newtonian amplitude.

4.2 Verification of the First-Order Entropy Variation

For weak perturbations, the identity

$$\Delta \log \det\{\}' A \approx \mathrm{Tr}((A^{-1})_{SS}\delta A_{SS}) \tag{20}$$

was evaluated numerically on the support S of the perturbation.

Across the tested parameter range, the ratio between the exact support-restricted log-determinant variation and the resolvent trace remained close to unity (typically within a few percent for small perturbations). This confirms that the weak-field expansion

$$\delta S_{\Pi} = \tfrac{1}{2}\mathrm{Tr}(A^{-1}\delta A) \tag{21}$$

is accurately realized in the discrete setting.

4.3 Spatial Support Threshold

The central result concerns the dependence of the Newtonian amplitude shift Δa on the spatial support of the perturbation.

For perturbations confined to a small region ($R = 1$), the entropy variation $\Delta \log \det\{\}' A$ is clearly non-zero, yet the fitted amplitude a remains essentially unchanged. The modification of the spectrum is therefore spectrally local and does not generate a measurable long-range effect.

For intermediate support ($R = 2$), the amplitude shift remains very small over the tested range.

In contrast, for perturbations extending over a larger region ($R = 3$ in the simulations), a coherent and monotonic modification of the Newtonian amplitude emerges. The amplitude shift Δa increases in magnitude as $-\Delta \log \det\{\}' A$ grows, as shown in Fig. 1.

These results demonstrate that entropy variation alone is not sufficient to produce a Newtonian interaction. A finite spatial coherence of the perturbation is required for the resolvent-mediated response to propagate to large distances.

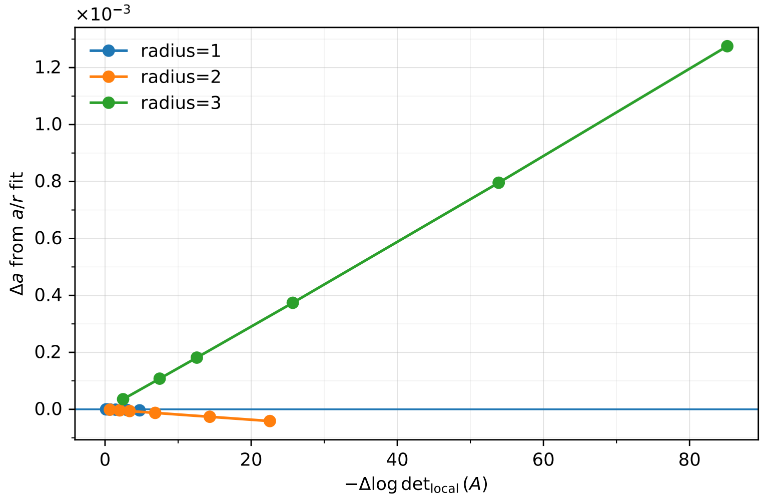

Fig. 1 Shift of the fitted Newtonian amplitude Δa as a function of $-\Delta \log \det\{\}' A$ for perturbations of increasing spatial radius R. Only sufficiently extended perturbations produce a macroscopic long-range response.

5 Continuum Limit and Scaling Considerations

5.1 Discrete-to-Continuum Correspondence

The numerical implementation employs a finite cubic lattice with Dirichlet boundary conditions. Let h denote the lattice spacing and L_h the corresponding graph Laplacian. In the limit $h \to 0$ with fixed physical domain size, the rescaled operator satisfies

$$h^{-2} L_h \longrightarrow -\nabla^2 \tag{22}$$

in the sense of operator convergence on suitable function spaces.

Accordingly, the discrete resolvent converges to the continuum Green operator,

$$(L_h + \varepsilon I)^{-1} \;\longrightarrow\; (-\nabla^2 + \varepsilon)^{-1}, \tag{23}$$

for fixed $\varepsilon > 0$. In three dimensions and for $\varepsilon \to 0$, the corresponding Green kernel asymptotically reproduces the Newtonian form

$$G(r) \sim \frac{1}{4\pi r}. \tag{24}$$

The discrete $1/r$ behavior observed numerically is therefore consistent with the continuum limit of Laplace-type operators.

5.2 Scaling of Entropy Variation

Under a rescaling of spatial coordinates $x \mapsto \lambda x$, the Laplacian transforms as

$$-\nabla^2 \mapsto \lambda^{-2}(-\nabla^2). \tag{25}$$

Eigenvalues scale accordingly as λ^{-2}, so that the logarithmic determinant transforms as

$$\log \det\{\}' A \mapsto \log \det\{\}' A - 2N \log \lambda, \tag{26}$$

where N denotes the number of positive modes below a fixed spectral cutoff.

Although the determinant itself depends on regularization, its first variation under localized perturbations,

$$\delta S_\Pi = \frac{1}{2} \mathrm{Tr}(A^{-1} \delta A), \tag{27}$$

is scale-covariant and governed by the Green kernel. The $1/r$ decay therefore follows from dimensional analysis in three spatial dimensions and is not an artifact of discretization.

5.3 Interpretation

The continuum scaling properties confirm that the Newtonian kernel arises from the elliptic nature of the operator rather than from lattice-specific effects. The discrete simulations reproduce the expected continuum behavior, while making explicit the role of finite spatial support in generating a macroscopic amplitude shift.

Within this framework, gravity appears as a long-range resolvent-mediated response of a projective entropy functional. The derivation is restricted to the weak-field, Newtonian regime and does not rely on additional geometric structure.

6 Discussion and Outlook

The analysis presented here establishes a minimal operator-theoretic mechanism by which a Newtonian interaction may emerge from the variation of a projective entropy functional. The key structural element is the resolvent identity

$$\delta S_\Pi = \frac{1}{2} \mathrm{Tr}(A^{-1} \delta A), \tag{28}$$

which makes explicit that entropy variation is mediated by the Green kernel of the underlying elliptic operator. In three spatial dimensions, this kernel exhibits a $1/r$ decay, thereby generating a Newtonian long-range response.

The numerical study confirms three essential features. First, the first-order approximation $\Delta \log \det\{\}' A \approx \mathrm{Tr}(A^{-1} \delta A)$ is accurately realized in the weak-perturbation regime. Second, the discrete Green function reproduces the expected $1/r$ behavior away from the source. Third, a finite spatial support of the perturbation is required for a macroscopic amplitude shift to appear. Entropy variation confined to a sufficiently small region remains spectrally local and does not induce a measurable long-range modification.

These observations indicate that long-range interaction arises from the interplay between spectral non-locality and spatial coherence of the perturbation. The mechanism does not assume a fundamental force law, nor does it rely on geometric curvature at the outset. Instead, gravitational behavior appears as a structural consequence of the resolvent of a three-dimensional Laplace-type operator.

The present work is restricted to the Newtonian regime. No relativistic dynamics, time dependence, or nonlinear backreaction have been introduced. Extending the mechanism to a fully covariant setting, or identifying the operator A with a dynamical geometric object, remains an open problem. It would also be of interest to analyze whether similar mechanisms in other spatial dimensions lead to modified long-range behavior, in accordance with the dimensional dependence of Green kernels.

In summary, the emergence of a Newtonian interaction can be understood as a resolvent-mediated response of a projective entropy functional. The result highlights the central role of spectral operators in generating long-range structure and suggests that gravity may admit an interpretation rooted in entropy variation rather than fundamental force postulates.

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